Experiment No.04

Name of Experiment: Study of Series and Parallel Circuit

Objectives:

a) To connect two types of connection circuits.

b) To measure the voltages and current across the resistors using the Multi-meter device.

c) To analyze the result and performance of the circuits.

Theory:

THE SERIES CIRCUIT

A series circuit or a "series-connected" circuit" is a circuit having just one current path. Thus, Fig 1(a) is an example of a "series circuit" in which a battery of constant potential difference V volts, and three resistances, are all connected "in series".

Since a series circuit has just one current path, it follows that all the components in a series carry the same current I, a fact evident from the inspection of Fig. 4(a).

The current I is assumed to be a flow of positive charge, and thus flows out of the positive terminal of the battery and around through the external circuit, reentering the battery at the end of the negative terminal. This is indicated by the arrows in Fig. 4(a).

In a series circuit, the TOTAL resistance, RT, that the battery sees is equal to the SUM of the individual resistance. Thus, in particular case of Fig. 4(a) the battery sees a total resistance, $R_T = R_1 + R_2 + R_3$ while in the general case of "n" resistances connected in series the battery sees a total resistance of..

$$R_T = R_1 + R_2 + R_3 \cdots R_n$$

By Ohm's Law, it follows that the current I in a series circuit is equal to:

$$I = \frac{V}{R_T} = \frac{V}{R_1 + R_2 + \cdots R_n}$$

In a series circuit, the applied voltage is equal to the sum of the voltage drops.

$$V_T = V_1 + V_2 + V_3$$

THE PARALLEL CIRCUIT

A parallel circuit is one in which the battery current divides into a number of "parallel paths". This shows up in Fig. 1(b) in which a battery, of constant V volts, delivers a current of I amperes to a load consisting of any numbers of n resistances connected "in parallel".

The currents in the individual resistances are called the "branch currents," and the battery current I is often called the "line current". From the inspection of Fig. 1 (b) we see that, in a parallel circuit, the battery current I is equal to the sum of the branch currents.

$$I = I_1 + I_2 + I_3 + \cdots + I_n$$

If the battery voltage V is applied equally to all n resistances, that is, the same voltage V is applied to all the parallel branches. Hence, by Ohm's law, the individual branch currents in Fig1(b) have the valves.

$$I_1 = \frac{V}{R_1}, \quad I_2 = \frac{V}{R_2}, \quad \cdots \quad I_n = \frac{V}{R_n}$$

Then we have,

$$I = V\left(\frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3} + \cdots + \frac{1}{R_n}\right)$$

Now let RT be the total resistance as seen by the battery in Fig 1(b). Then, by Ohm's law, it has to be true that:

$$I = \frac{V}{R_T}$$

CIRCUIT DIAGRAM

Series Circuit

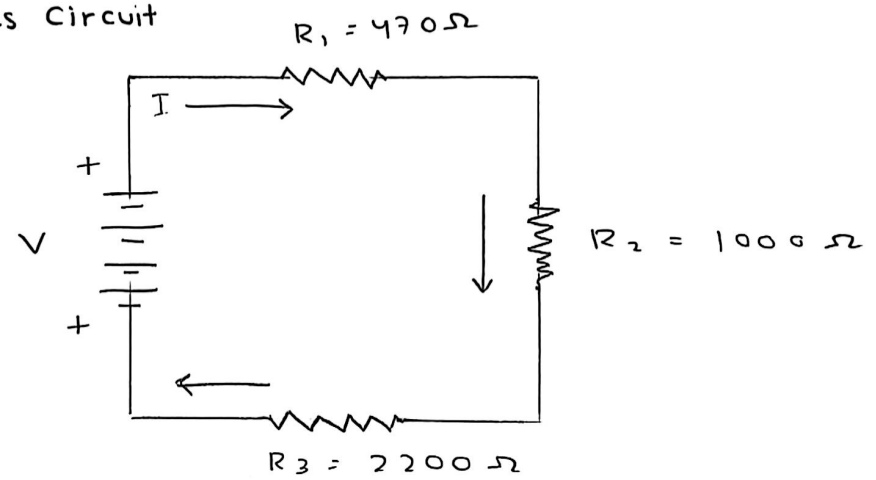


Figure No 1(a): Series Circuit

Parallel Circuit

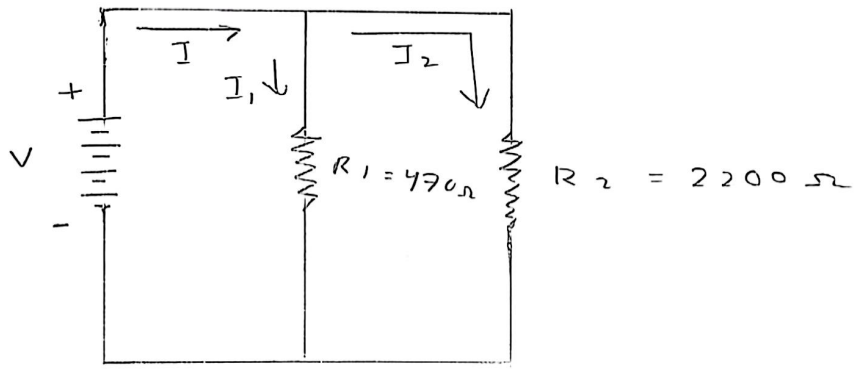


Figure No. 1(b): Parallel Circuit

Apparatus

- DC Power Supply
- Breadboard
- Resistors of various values
- voltmeter
- ammeter
- connecting wires.

Procedure

a) Connect all components on the breadboard according to the circuit.

b) Turn on the DC power supply and set it to 4V.

c) Measure the voltage across each resistor (series) and the current through each branch (parallel) using the voltmeter and ammeter respectively.

d) Repeat the above steps for supply voltage of 5V and 6V.

Table Data 1:

For Series Circuit,

obs	V	R1 (Ω)	R2 (Ω)	R3 (Ω)	V1(V)		V2(V)		V3(V)		Rs(Ω)	
					Cal	Mea	Cal	Mea	Cal	Mea	Cal	Mea
1	4	470	1000	2200	0.51	0.52	1.08	1.12	2.39	2.42	3670	3.55
2	5				0.64	0.64	1.36	1.38	2.99	3.03		
3	6				0.76	0.77	1.63	1.66	3.59	3.64		

Table Data 2:

For Parallel Circuit,

obs	V	R1(Ω)	R2(Ω)	I1 (mA)		I2 (mA)		RP (Ω)	
				cal	mea	cal	mea	cal	mea
1	4	470	2200	8.2	8.6	1.7	1.8	387.26	0.379
2	5			9.8	10.8	2.1	2.2		
3	6			12.3	13.0	2.6	2.7		

Zakir 01.10.26

Submission: 08.10.26

Calculations

A Series Circuit,

$$R_s = R_1 + R_2 + R_3$$

$$= 470 + 1000 + 2200$$

$$= 3670\ \Omega$$

OBS 1

$$V_1 = \frac{V \times R_1}{R_1 + R_2 + R_3} = \frac{4 \times 470}{3670} = 0.51\ V$$

$$V_2 = \frac{V \times R_2}{R_1 + R_2 + R_3} = \frac{4 \times 1000}{3670} = 1.09\ V$$

$$V_3 = \frac{V \times R_3}{R_1 + R_2 + R_3} = \frac{4 \times 2200}{3670} = 2.40\ V$$

OBS 2

$$V_1 = \frac{V \times R_1}{R_1 + R_2 + R_3} = \frac{5 \times 470}{3670} = 0.64V$$

$$V_2 = \frac{V \times R_2}{R_1 + R_2 + R_3} = \frac{5 \times 100}{3670} = 1.36V$$

$$V_3 = \frac{V \times R_3}{R_1 + R_2 + R_3} = \frac{5 \times 2200}{3670} = 3.00$$

OBS 3 (V = 6V)

$$V_1 = \frac{V \times R_1}{R_1 + R_2 + R_3} = \frac{6 \times 470}{3670} = 0.77\,V$$

$$V_2 = \frac{V \times R_2}{R_1 + R_2 + R_3} = \frac{6 \times 1000}{3670} = 1.63\,V$$

$$V_3 = \frac{V \times R_3}{R_1 + R_2 + R_3} = \frac{6 \times 2200}{3670} = 3.60\,V\,\Omega$$

$$R_S = \frac{V}{I} = (V_1 + V_2 + V_3)\,I = 3550\,\Omega$$

A PARALLEL CIRCUIT

$$R_p = \frac{R_1 \times R_2}{R_1 + R_2}$$

$$= \frac{470 \times 2200}{470 + 2200}$$

$$= 387.27\,\Omega$$

OBS 1 (V = 4V)

$$I = \frac{V}{R} = \frac{4}{387.27} = 10.33\,A$$

$$I_1 = \frac{I \times R_2}{R_1 + R_2} = \frac{10.33 \times 2200}{2670} = 8.51\,mA$$

$$I_2 = \frac{I \times R1}{R_1 + R_2} = \frac{10.33 \times 470}{2670} = 1.82\,mA$$

OBS 2 (V = 5V)

$$I = \frac{V}{R_p} = \frac{5}{387.27} = 12.91\ mA$$

$$I_1 = \frac{I \times R_2}{(R_1+R_2+R_3)} = \frac{12.91 \times 2200}{2670} = 10.64\ mA$$

$$I_2 = \frac{I \times R}{(R_1+R_2+R_3)} = \frac{12.91 \times 470}{2670} = 2.27\ mA$$

OBS 3: (V = 6V)

$$I = \frac{V}{R_p} = \frac{6}{387.27} = 15.49\ mA$$

$$I_1 = \frac{I \times R_2}{R_1+R_2} = \frac{15.49 \times 2200}{2670} = 12.77\ mA$$

$$I_2 = \frac{I \times R_1}{R_1+R_2} = \frac{15.49 \times 470}{2670} = 2.73\ mA$$

$$R_p = \frac{V}{I} = \frac{V}{(I_1+I_2)} = 379\ \Omega$$

RESULTS

A Series Circuit

a) OBS 1: Measured Value of Resistor = 3550 Ω
Calculated Value of Resistor = 3670 Ω

b) OBS 2: Measured Value of Resistor = 3550 Ω
Calculated Value of Resistor = 3670 Ω

c) OBS 3: Measured Value of Resistor = 3550 Ω
Calculated Value of Resistor = 3670 Ω

A Parallel Circuit

a) OBS 1: Measured Value of Resistor = 379 Ω
Calculated Value of Resistor = 387.27 Ω

b) OBS 2: Measured Value of Resistor = 379 Ω
Calculated Value of Resistor = 387.27 Ω

c) OBS 3: Measured Value of Resistor = 379 Ω
Calculated Value of Resistor = 387.27 Ω

Percentage of Difference $\% R_p = \frac{387.27 - 379}{387.27 \times 100} = 2.13\%$

DISCUSSION

The observed values and calculated values are nearly the same. The difference between tho measured and calculated valves are due to instrumental and observational errors.

N.B : Fluctuations in temperature, humidity, and other environmental conditions can affect the performance of components.

Conclusion:

In a series circuit the resistances add and the supply voltage divides among the resistors. In a parallel circuit, the voltage is the same across the resistors' in all branches. The branch current adds up to the Tinc currend and $R_p = \frac{R_1 R_2}{R_1 + R_2}$.

The measured and calculated values agreed closely, so both circuits were verified.